Research Statement

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Research Experience

My research addresses the **integration of AI/ML into electronic and electrical engineering**, with a focus on using deep learning to accelerate the multi-physics computational workflows that underpin electric machine and power-conversion device design. I hold a PhD in Electrical Engineering from Southeast University (SEU, 2025) and am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham. I have authored 22 journal papers (17 in IEEE Transactions, ten as first author), 18 conference papers and 9 patents, recognised by the 2025 IEEE TEC Electric Vehicle Design Competition (1st Prize) and the 2025 *IEEE Trans. Transportation Electrification* Prize Paper Award (2nd Prize).

AI/ML to accelerate traditional computational workflows. My ICEM 2026 paper [C1] proposes a deep-learning magnetic-field surrogate with **implicit physics-knowledge embedding**: rather than mapping design parameters to scalar performance figures, a residual network with a Fourier coordinate-encoding layer learns the full magnetic vector potential field over a 15-dimensional design space, so that torque, flux linkage and iron loss follow from the same post-processing used in finite-element analysis. On unseen validation designs the model reaches $R^2 = 0.9996$ for average torque and 0.9730 for iron loss (against 0.9533 and 0.9410 for Kriging), with inference time reduced by **more than 94%**, and physical structure such as field symmetry emerging without being prescribed. Training runs on a workstation GPU while inference runs on low-cost consumer hardware, turning a CPU-bound FE workflow into a **GPU-accelerated** one. The same philosophy of physics-grounded acceleration runs through my published optimisation work [J6], where index models built from non-linear magnetic circuits cut a 67-hour FE-based multi-objective optimisation to 2.3 hours while finding better Pareto designs, validated on a prototype machine.

Scientific computing foundation. These AI methods stand on an **in-house Python finite-element code package** I develop, covering electromagnetic (including moving-band transient), structural-modal and mechanical-stress analyses, plus an object-oriented lumped-parameter thermal solver. Owning the solver stack means training data, physics constraints and AI models can be co-designed end to end, which is the practical prerequisite for AI-native simulation research.

Multi-physics and interdisciplinary modelling. My *IEEE Trans. Transportation Electrification* paper [J10] develops a high-fidelity lumped-parameter thermal model with fluid-circuit coupling for oil-cooled hairpin-winding machines, resolving every conductor in every slot and the coolant mass-transfer transient. Validated against 48 thermocouples on a dynamometer-tested prototype, it predicts temperatures to within 5 °C in seconds of computation, against the >120 CPU-hours reported for equivalent CFD, and its calibration loop itself uses a neural-network surrogate. This work reflects the interdisciplinary span of my research across electromagnetic, thermal, fluid and mechanical physics.

Manufacturing-aware design for sustainable transport. My prize-winning paper on *n*-layer flat-wire winding design [J5] generalises hairpin winding layouts from even to arbitrary layer numbers, easing the pairing of torque-speed envelope and DC bus voltage while reducing pin types and manufacturing complexity. Through a multi-year collaboration with Inovance, a world-leading EV traction-machine supplier, I carried two oil-cooled PM traction machines (18,000 rpm, 113 kW; 20,000 rpm, 270 kW) through the full industrial cycle from multi-physics design and manufacturing-drawing release to AVL dynamometer characterisation. **Data-centric design** is embedded in this practice: structured FE datasets generated by Latin hypercube sampling (45,000 candidate designs in [C1]) and a patented dual-model strategy for building high-precision machine performance databases [P3] supply the data infrastructure on which the AI models train.

Five-Year Research Plan

At Bath I would build a research programme on **AI-native simulation and design of electromagnetic energy-conversion devices**, aligned with **The Foundry: Centre for Digital, Manufacturing & Design** and

contributing to sustainable engineering through transport electrification. The programme has three connected themes.

Theme 1: General AI/ML modelling for electrical machine analysis. The ICEM 2026 surrogate is an initial attempt at involving AI/ML in finite-element analysis for electrical machines; the disruptive step is to generalise it. I will extend the approach toward a **general, GPU-accelerated learned simulator** covering variable material topology (geometry-conditioned encoders over different machine classes), eddy-current and PWM-driven excitation conditions, and coupled electromagnetic-thermal fields, drawing on operator-learning and physics-informed methods (Fourier neural operators, DeepONet, PINN-style constraints). Because the models are differentiable, they unlock gradient-based topology optimisation, inverse design from performance specifications, and sensitivity analysis of manufacturing tolerances, capabilities that conventional FE workflows cannot offer. Years 1 to 2 deliver the eddy-current and PWM-excitation extensions and a benchmark dataset; years 3 to 5 target cross-device generalisation and differentiable design demonstrations.

Theme 2: Physics-grounded digital twins and data-centric design. Building on the real-time thermal model [J10] and the field surrogate [C1], I will develop digital twins that run alongside drive systems for thermally aware control, condition monitoring and rapid what-if analysis, supported by curated design databases [P3]. This theme connects naturally to data-centric and big-data approaches in engineering design and to industrial deployment.

Theme 3: Manufacturing-aware machine design for sustainable transport. Extending the *n*-layer winding methodology [J5] and tolerance-aware multi-physics design into a framework where manufacturability, cost and sustainability enter the optimisation loop directly, in partnership with drive-system and powertrain manufacturers, which is where the programme meets The Foundry's agenda of digital tools for a sustainable manufacturing sector.

Funding and collaboration. I would pursue an **EPSRC New Investigator Award** within the first eighteen months to establish Theme 1, followed by standard EPSRC responsive-mode and **Innovate UK** industrially co-funded projects building on my prototype delivery record, with Advanced Propulsion Centre, Aerospace Technology Institute and Horizon Europe consortium bids as the group grows. My current post within the multi-institution AIMD programme (Nottingham, Bristol, Newcastle) has given me working knowledge of how large collaborative UK programmes are structured and delivered, and my co-author network already includes the University of Bath through joint work with colleagues associated with IAAPS. The programme will generate PhD projects and computational teaching material linking directly to the Department's applied computer systems engineering and AI curriculum.

Paper identifiers [J#], [C#], [P#] refer to the publication list in the attached CV.